

1. The number 3.14063 written correct to 3 decimal places is
- (A) 3.140
(B) 3.141
(C) 3.146
(D) 3.150
2. Express 0.12 as a fraction in its lowest terms.
- (A) $\frac{1}{9}$
(B) $\frac{3}{5}$
(C) $\frac{1}{8}$
(D) $\frac{3}{25}$
3. In scientific notation, 170.04 is written as
- (A) 0.17004×10^3
(B) 1.7004×10^2
(C) 17.004×10^1
(D) 1.7004×10^{-1}
4. There are 40 students in a class. Girls make up 60% of the class. 25% of the girls wear glasses. How many girls in the class wear glasses?
- (A) 6
(B) 8
(C) 10
(D) 15
5. If \$560 is shared in the ratio 2 : 3 : 9, then the difference between the largest and the smallest shares is
- (A) \$ 80
(B) \$240
(C) \$280
(D) \$360
6. If 60% of a number is 90, what is the number?
- (A) 30
(B) 54
(C) 150
(D) 180
7. The first three common multiples of 3, 4 and 6 are:
- (A) 12, 24, 36
(B) 6, 8, 12
(C) 3, 4, 6
(D) 0, 1, 2
8. The number 301 can be written as
- (A) $3 \times 10^2 + 1$
(B) $3 \times 10^3 + 1$
(C) $3 \times 10^2 + 1 \times 10$
(D) $3 \times 10^3 + 1 \times 10$
9. The value of the digit 5 in the number 537 is
- (A) $\frac{5}{500}$
(B) 50
(C) 500
(D) 5000
10. By the distributive law $49 \times 17 + 49 \times 3 =$
- (A) $52 + 66$
(B) 52×66
(C) $49 + 20$
(D) 49×20

GO ON TO THE NEXT PAGE

11. Which of the following sets is defined by $\{x \in \mathbb{Z} : -2 \leq x \leq 4\}$?
- (A) $\{0, 1, 2, 3, 4\}$
 (B) $\{1, 2, 3, 4\}$
 (C) $\{-1, 0, 1, 2, 3\}$
 (D) $\{-2, -1, 0, 1, 2, 3, 4\}$
12. In a class of 32 students, 17 study Music and 20 study Art. What is the LEAST number of students who are studying BOTH Music and Art?
- (A) 3
 (B) 5
 (C) 12
 (D) 15
15. A dress which costs \$180 is being sold at a discount of 10%. The amount of the discount is
- (A) \$ 1.80
 (B) \$ 10.00
 (C) \$ 18.00
 (D) \$170.00
16. A man's annual salary is \$45 000. His tax-free allowances total \$13 000. He has to pay a tax of 35% on his taxable income. The tax payable is
- (A) \$ 4 550.00
 (B) \$11 200.00
 (C) \$15 750.00
 (D) \$20 300.00

Item 13 refers to the following information.

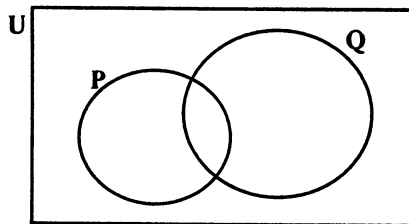
$P = \{\text{prime numbers}\}$

$Q = \{\text{odd numbers}\}$

$R = \{\text{even numbers}\}$

13. Which of the following sets is empty?
- (A) $P \cap R$
 (B) $Q \cap R$
 (C) $P \cap Q$
 (D) $P \cup Q$
17. If TT\$6.00 is equivalent to US\$1.00, then TT\$15.00 in U.S. dollars is
- (A) \$0.25
 (B) \$0.40
 (C) \$2.50
 (D) \$4.00
18. If \$7000 is borrowed at the rate of 5% per annum for 3 years, the simple interest is
- (A) \$ 105
 (B) \$ 210
 (C) \$ 370
 (D) \$1050

Item 14 refers to the Venn diagram below.



14. In the Venn diagram, $n(P) = 5$, $n(Q) = 9$ and $n(P \cup Q) = 10$. What is $n(P \cap Q)$?
- (A) 4
 (B) 6
 (C) 14
 (D) 24
19. How much does a customer pay for an article marked at \$50.00 if a sales tax of 6% is charged?
- (A) \$44.00
 (B) \$47.00
 (C) \$53.00
 (D) \$56.00

20. Tracy bought a pen for \$60 and sold it to gain 20% on her cost price. How much money did she gain?
- (A) \$12
(B) \$40
(C) \$72
(D) \$80
21. A salesman is paid 5% of his sales as commission. His sales for last month were \$2 020. How much commission was he paid?
- (A) \$ 11.00
(B) \$ 20.20
(C) \$101.00
(D) \$110.00
22. A loan of \$8 000 was paid back in 2 years in monthly payments of \$400.00. The percentage interest on the loan was
- (A) 5%
(B) $8\frac{1}{3}\%$
(C) $16\frac{1}{3}\%$
(D) 20%
23. If $r * s = s^r$, then $3 * 2 =$
- (A) 8
(B) 9
(C) 12
(D) 27
24. $\frac{4}{5x} + \frac{2}{5x} =$
- (A) $\frac{6}{25x}$
(B) $\frac{8}{25x}$
(C) $\frac{6}{10x}$
(D) $\frac{6}{5x}$
25. $5(x + y) - 3(x - y) =$
- (A) $2x$
(B) $2x + 2y$
(C) $2x + 8y$
(D) $8x + 8y$
26. If $x = -2, y = 3, t = 2$, then $\left(\frac{x}{y}\right)^t =$
- (A) $-\frac{4}{9}$
(B) $\frac{4}{9}$
(C) $\frac{4}{3}$
(D) $\frac{9}{4}$
27. $3x^2 \times 2x^3 =$
- (A) $6x^5$
(B) $5x^5$
(C) $5x^6$
(D) $72x^5$

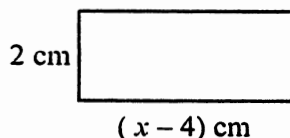
28. Given $2x + 3 \geq 9$, the range of values of x is

(A) $x > 6$
(B) $x \geq 6$
(C) $x > 3$
(D) $x \geq 3$

29. If $5x - 26 = x + 50$, then the value of x is

(A) -12
(B) -6
(C) 6
(D) 19

Item 30 refers to the following diagram.



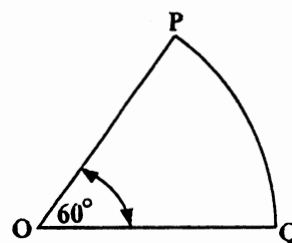
30. The area of the rectangle, in cm^2 , is x^2 . Which of the following equations may be used to find the value of x ?

(A) $x^2 = 2(x - 4)$
(B) $x^2 = (x - 2)(x - 4)$
(C) $x^2 = 2(x - 4)(x - 2)$
(D) $x^2 = (x - 4)(x + 2)$

31. If $3 + \frac{2}{x} = 1$, then the value of x is

(A) -1
(B) $\frac{1}{5}$
(C) $\frac{1}{2}$
(D) 5

Item 32 refers to the figure below.



32. The figure above, not drawn to scale, shows a sector of a circle centre O. The length of the minor arc PQ is 8 cm. What would be the length of the circumference of the circle?

(A) 16 cm
(B) 24 cm
(C) 48 cm
(D) 64 cm

33. The volume of a cube whose edge is 6 cm long is

(A) 18 cm^3
(B) 36 cm^3
(C) 72 cm^3
(D) 216 cm^3

34. 2500 millimetres expressed in metres is

(A) 0.25
(B) 2.5
(C) 25
(D) 250

35. Tom leaves town P at 0600 hrs and drives to town Q, which is 595 km away. He arrives in town Q at 1300 hrs the same day. Tom's average speed was

(A) 70 km/h
(B) 75 km/h
(C) 85 km/h
(D) 90 km/h

GO ON TO THE NEXT PAGE

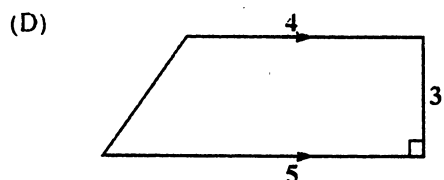
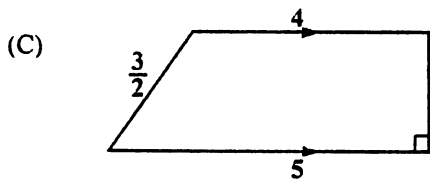
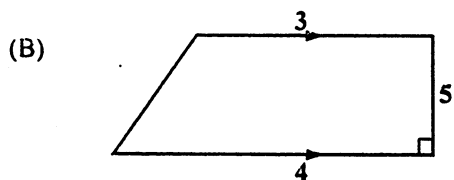
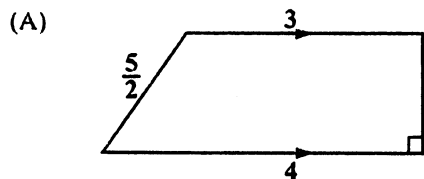
36. On leaving Trinidad, the time on a pilot's watch was 23:00 hrs. When he arrived at his destination, in the same time zone, on the following day his watch showed 03:00 hrs. How many hours did the flight take?

(A) 4
(B) 20
(C) 26
(D) 52

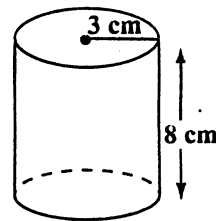
37. Fifty guests were each served 2 glasses of champagne. Each glass held 150 millilitres. How many litres of champagne were served?

(A) 0.15
(B) 1.5
(C) 15
(D) 150

38. Which of the figures below, **not drawn to scale**, has an area equal to $\frac{1}{2}(3 + 4) \times 5$ square units?



Item 39 refers to the diagram below.



39. The diagram, **not drawn to scale**, shows a cylinder of radius 3 cm and height of 8 cm. The volume is

(A) $12\pi \text{ cm}^3$
(B) $48\pi \text{ cm}^3$
(C) $72\pi \text{ cm}^3$
(D) $192\pi \text{ cm}^3$

40. The area of a triangle is 30 cm^2 and its base is 10 cm. What is the perpendicular height of the triangle?

(A) 6 cm
(B) 12 cm
(C) 13 cm
(D) 17 cm

41. The heights, in cm, of ten students are 150, 152, 153, 155, 155, 156, 158, 160, 165, 170. The **range** is

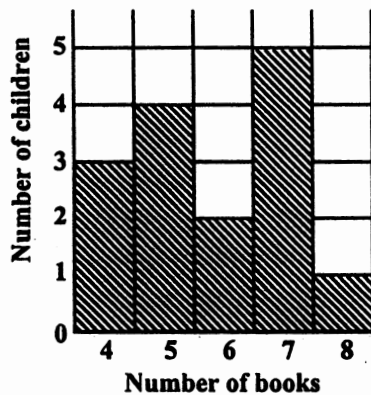
(A) 5
(B) 20
(C) 150
(D) 155

42. A bag contains 60 marbles of different colours but similar in size and mass. The probability of randomly choosing a red marble is $\frac{5}{12}$. How many red marbles does the bag contain?

(A) 5
(B) 12
(C) 17
(D) 25

GO ON TO THE NEXT PAGE

Item 43 refers to the following bar chart.



43. The bar chart above shows the number of books read by the children who took part in a survey.

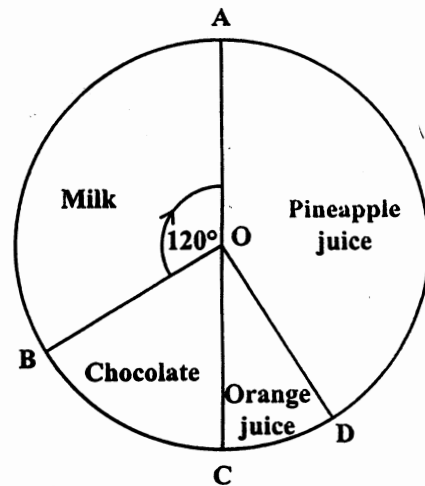
How many children took part in the survey?

- (A) 5
(B) 15
(C) 75
(D) 87

44. If the mean of the four numbers 4, 8, x and 12 is 10, then x is

- (A) 4
(B) 10
(C) 12
(D) 16

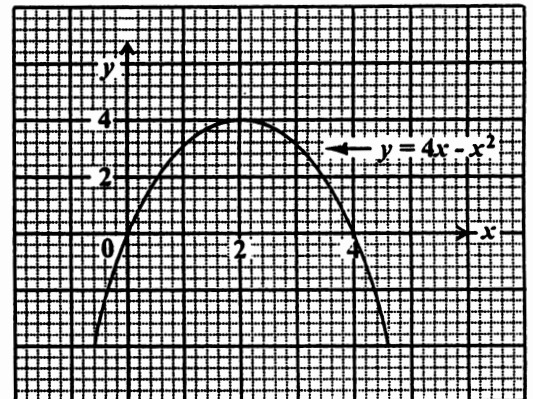
Item 45 refers to the following diagram.



45. The pie-chart above shows the preference in drinks of a group of students. If 12 students prefer chocolate, then the TOTAL number of students is

- (A) 48
(B) 72
(C) 180
(D) 360

Item 46 refers to the graph below.

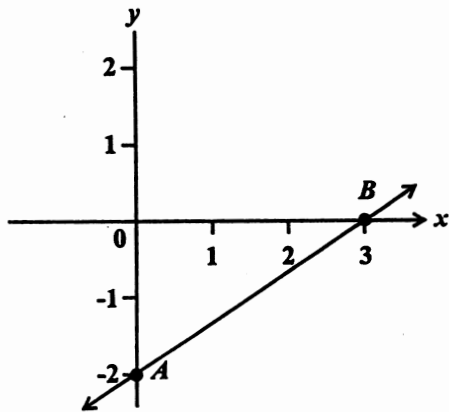


46. The maximum point of $y = 4x - x^2$ is

- (A) (4, 2)
(B) (2, 4)
(C) (0, 4)
(D) (0, 0)

GO ON TO THE NEXT PAGE

Item 47 refers to the graph below.



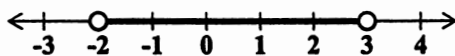
47. The straight line AB cuts the Y axis at

- (A) (0, 3)
- (B) (0, 2)
- (C) (3, -2)
- (D) (0, -2)

48. Which of the following sets is represented by the relation $f: x \rightarrow x^2 + 3$?

- (A) $\{(0, 3), (1, 4), (2, 7), (3, 12)\}$
- (B) $\{(0, 3), (1, 5), (2, 7), (3, 9)\}$
- (C) $\{(0, 3), (1, 4), (2, 5), (3, 6)\}$
- (D) $\{(0, 3), (1, 1), (2, 4), (3, 9)\}$

Item 49 refers to the diagram below.



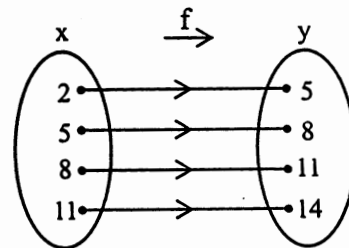
49. The inequality above is defined by

- (A) $-2 \leq x < 3$
- (B) $-2 < x \leq 3$
- (C) $-2 \leq x \leq 3$
- (D) $-2 < x < 3$

50. If $h(x) = \frac{3x-2}{5}$, then $h(-6) =$

- (A) -4
- (B) $-\frac{16}{5}$
- (C) $\frac{16}{5}$
- (D) 4

Item 51 refers to the arrow diagram below, which shows a function.



51. Which of the following BEST describes the function?

- (A) $y = x$
- (B) $x = y + 3$
- (C) $f(x) = x + 3$
- (D) $f(x) = y + 3$

52. A boat was travelling on a bearing of 270° . In what direction was it travelling?

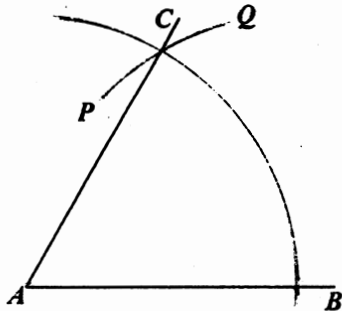
- (A) East
- (B) West
- (C) North
- (D) South

53. If the sum of the interior angles of a polygon is 4 right angles, the polygon is a

- (A) triangle
- (B) hexagon
- (C) pentagon
- (D) quadrilateral

GO ON TO THE NEXT PAGE

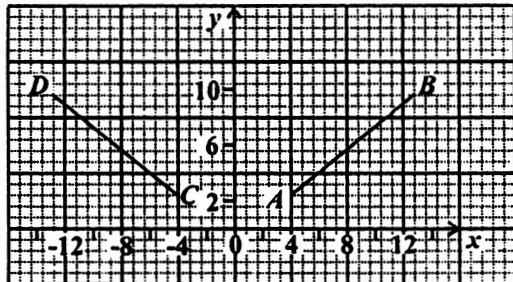
Item 54 refers to the diagram below of a construction. With centre A, an arc BC is drawn. With centre B, and the same radius, the arc PCQ is drawn.



54. What is the measure of $\angle BAC$?

- (A) 30°
- (B) 45°
- (C) 60°
- (D) 75°

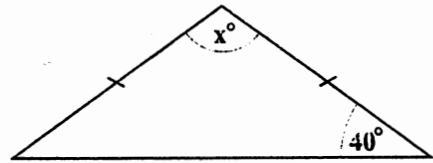
Item 55 refers to the following graph.



55. In the figure above, the line CD is the image of AB after

- (A) a rotation through 90° centre O
- (B) a reflection in the y -axis
- (C) a translation by vector $\begin{pmatrix} -4 \\ -8 \end{pmatrix}$
- (D) an enlargement of scale factor -1

Item 56 refers to the isosceles triangle below.



56. In the triangle, the value of x is

- (A) 40°
- (B) 80°
- (C) 100°
- (D) 140°

57. The image of a point $P(-2, 3)$ under a translation $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ is

- (A) $(-6, 12)$
- (B) $(-5, -1)$
- (C) $(5, 1)$
- (D) $(1, 7)$

58. A ladder 5 metres long is leaning against a vertical wall. The foot of the ladder is 3 metres away from the wall. How far up the wall does the ladder reach?

- (A) 4 m
- (B) 6 m
- (C) 8 m
- (D) 15 m

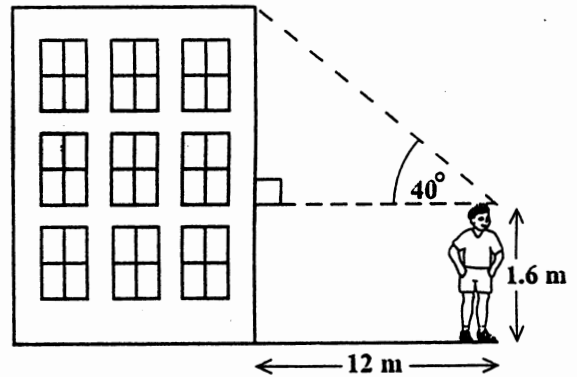
GO ON TO THE NEXT PAGE

59. In a triangle ABC, angle $A = x^\circ$ and angle $B = 2x^\circ$. What is the size of angle C?

- (A) 45°
- (B) 60°
- (C) $\left(\frac{180}{3x}\right)^\circ$
- (D) $(180 - 3x)^\circ$

Item 60 refers to the diagram of a building below.

A boy stands 12 metres from the foot of the building and observes the angle of elevation of the top of the building.



60. The height of the building is approximately

- (A) $12 \tan 40^\circ$
- (B) $1.6 + 12 \sin 40^\circ$
- (C) $1.6 + 12 \cos 40^\circ$
- (D) $1.6 + 12 \tan 40^\circ$

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.